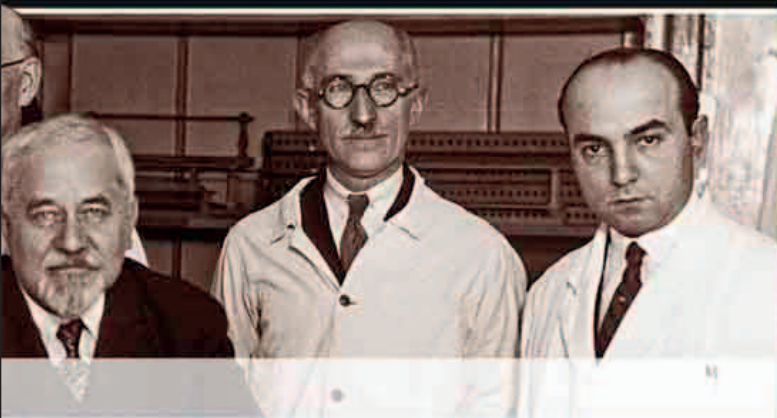
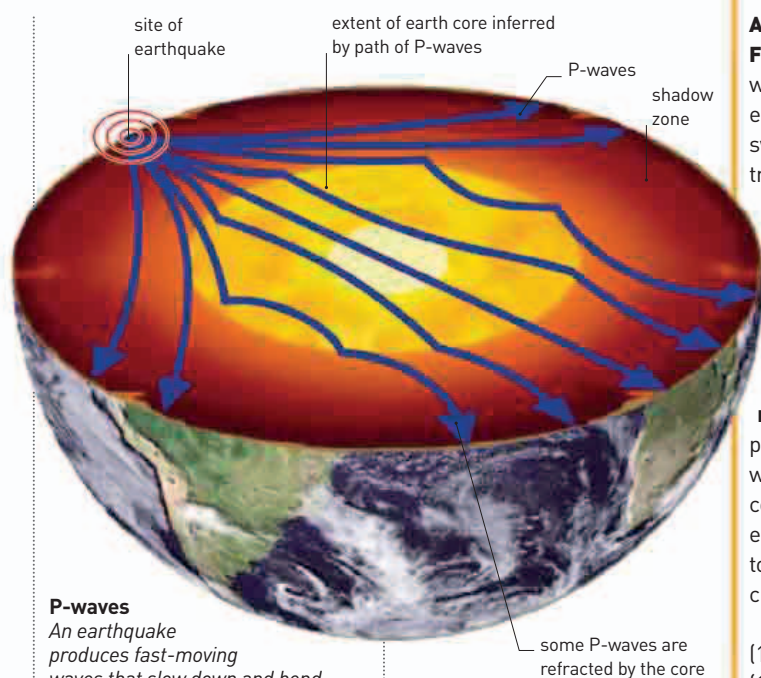




By 1907, Nobel laureate Ivan Pavlov, shown here with his staff and one of the dogs, had received much acclaim for his experiments with reflex responses in animals.



P-waves

An earthquake produces fast-moving waves that slow down and bend as they pass through Earth's core, creating a "shadow" zone where a seismograph cannot detect them.

out for TB. Calmette and Guérin wanted to use a **bovine (cattle-borne) form of TB to develop a vaccine** against the human form of TB. They set up cultures of bovine TB on potato slices that

had been soaked with ox gall (obtained from cow's liver) and set out to continue the culture line until it was safe to use. It was 10 years before their **BCG (Bacille Calmette Guérin) vaccine** was ready for use on animals—it was not tried on humans until 1921.

AMERICAN INVENTOR LEE DE FOREST PATENTED THE TRIODE,

which could be used to amplify electrical signals and act as a switch. Until the invention of the transistor years later (see 1947), diode and triode valves were used in circuits in radios, televisions, and computers.

The Belgian-born chemist **Leo Baekeland** (1863–1944) produced the **first plastic made from synthetic materials**. Baekeland's new product, later named Bakelite, was heat-resistant and non-conductive. It was used for electrical insulation as well as to make domestic utensils and children's toys.

French filmmakers **Auguste (1862–1954) and Louis Lumière (1864–1948)** began marketing the **first commercial color photographic method**. This **autochrome** (self-coloring) process involved using a negative plate that was coated with transparent grains of dye-colored starch, which filtered light before it hit a layer of photographic emulsion.

Following J.J. Thomson's demonstration that electrons were particles (see 1896), American physicist **Robert Millikan** (1868–1953) began experiments to calculate their electrical charge. He found that the charge on falling droplets of

“THE MATERIAL OF 1,000 USES.”

Leo Baekeland, on Bakelite

oil was always a multiple of a tiny value: **the charge on a single electron**. Millikan published his first results in 1910.

Following the discovery that radioactive materials decay at fixed rates (see 1901), American scientist **Bertram Boltwood** (1849–1936) used this to **calculate the age of rocks**. He found

that ores of uranium (uraninite) contained a proportion of lead, and the older the rock, the more lead it contained. Lead was a product of a known rate of uranium decay, so it accumulated over time. The study of radiometric dating was

Early color photograph

This autochrome image shows Doug, niece of the Lumière brothers, out in her pram with her nurse. It was taken between 1906 and 1912.



later taken up by British geologist Arthur Holmes (see 1913).

The German chemist **Emil Fischer** (1852–1919) succeeded in **linking together the building blocks of proteins**. These units—called **amino acids**—came in a variety of types. Fischer identified many of them and showed how they bonded in protein chains. His work was the foundation of protein chemistry.

Russian physiologist **Ivan Pavlov** (1849–1936) converted his laboratory so that he could concentrate on his **study of animal behavior**. Using dogs, Pavlov had shown that animals could be taught to salivate when they heard a ringing bell, known as the **conditioned, or learned, reflex** (see 1889).

9,800°F

THE TEMPERATURE OF EARTH'S INNER CORE

Swedish physicist **Theodor Svedberg** describes experimental tests for Einstein's theory of Brownian motion

Dutch biologist **Nicolaas Sönnigen** describes the first known **methane-producing bacteria**

November 4 Alois Alzheimer produces public report on the brain of a dead dementia patient

December 24 Reginald Fessenden transmits the first AM radio entertainment broadcast

February 18 US Patent Office registers the triode for Lee De Forest

February Bertram Boltwood describes radiometric dating of minerals

Paul Ehrlich develops arsenic-based drug for treating sleeping sickness

July Leo Baekeland files his method for making a synthetic plastic with US Patent Office

June Auguste and Louis Lumière start commercial production of **Autochrome Lumière**

Norwegian biochemists **Axel Holst and Alfred Frølich** show that scurvy is caused by nutrient deficiency—later identified as vitamin C

October 18 Emil Fischer describes bonding amino acids together in chains

November 4 Ivan Pavlov is elected Academician of the Russian Academy of Sciences

Late 1907 Robert Millikan and Harvey Fletcher devise the oil-drop experiment to measure the charge on an electron